NMEC595 - Research Methodology

Lecture 4: Research Design

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Lecture Outline

1. Meaning of Research Design
2. Need for Research Design
3. Features of Good Design
4. Important Concepts
5. Types of Research Studies
6. Experimental Design Types
7. Design by Research Purpose
8. Research Design Components
9. Planning Considerations

Research Problem: A brief Recap

- The first step in research is to identify and define the problem
- A problem should include five essential elements: individual or group, objectives, alternative options, uncertainty, and context
- Selection requires attention to feasibility, significance, and researcher's background
- Formulation involves two stages: first, thorough understanding; second, clear operational rephrasing
- Defining the problem is the most critical stage of the research process
- Use an iterative process: Refine the problem statement through repeated review for greater specificity
- A clear problem definition supports effective planning and execution of research

What is Research Design?

Research Design means:

- A **plan** for how to collect, measure, and look at information
- Main Parts:
  1. **A clear question** or problem you want to solve
  2. **Steps and methods** for finding information
  3. **How you will organize** and understand the information you collect
  4. **Who or what** you will study (the group or population)

Why is Research Design Important?

Research Design is Important Because:

- **Keeps things organized**: Helps the research go step by step in an orderly way
- **Saves time and resources**: Helps you get the most information without wasting time or money
- **Helps you plan ahead**: Lets you decide on your methods before you start
- **Makes results trustworthy**: Gives you results you can trust and believe in
- **Avoids mistakes**: Stops you from making wrong conclusions or wasting effort

Analogy: Just like a blueprint helps build a house the right way, a research design helps you do your research the right way.

What Makes a Good Research Design?

A Good Research Design Should Be:

- **Flexible**: Can adjust to look at different parts of the problem
- **Suitable**: Matches the goals and situation of your research
- **Efficient**: Gets the most useful information using the least time and resources
- **Trustworthy**: Reduces errors and gives reliable results
- **Cost-effective**: Fits within your time and budget limits

Note: The best research design depends on your research goals, the type of problem, what resources you have, and the skills of your team.

Key Ideas in Research Design (I)

Types of Variables:

- **Dependent Variable**: The result or outcome you are measuring
- **Independent Variable**: The factor you think causes a change
- **Continuous Variable**: Can be any value within a range (like height or weight)
- **Discrete Variable**: Can only be whole numbers (like number of children)

Example: If you want to see how age affects height, then height is what you measure (dependent variable) and age is what might cause the change (independent variable).

Key Ideas in Research Design (II)

Control and Extra Variables:

- **Extraneous Variable**: Something not part of your main study, but it can still affect your results
- **Control**: Steps you take to reduce the effect of these extra variables on what you are measuring
- **Confounded Relationship**: When your results are mixed up because extra variables are affecting them

Hypothesis-Testing Research: Purpose and Approaches

Purpose: Evaluate whether a proposed explanation (hypothesis) about a relationship between variables is supported by evidence. This process aims to test claims of causation or association using systematic observation and analysis.

Research Approaches

- Hypothesis-testing research can be conducted using two main approaches:
  - **Experimental Research**: The researcher deliberately alters one or more independent variables to observe effects on dependent variables.
  - **Non-Experimental (Observational) Research**: The researcher observes phenomena as they occur naturally, without intervention.

Importance of Randomization in Experimental Design

Randomization is the process of assigning subjects or experimental units to groups (such as control and experimental groups) using chance methods, such as random number tables or computer algorithms.

Why is Randomization Important?

- **Reduces selection bias:** Ensures that each participant has an equal chance of being assigned to any group, minimizing systematic differences.
- **Controls confounding variables:** Distributes both known and unknown factors evenly across groups, making the groups comparable.
- **Supports valid inference:** Allows for the use of statistical tests that rely on probability theory, increasing the credibility and generalizability of results.

Key Point: Randomization is fundamental for drawing reliable and unbiased conclusions about cause and effect in experimental research.

Formal and Informal Experimental Designs

Informal Experimental Designs

- Lack full randomization or rigorous control.
- Often used in preliminary studies or when strict control is not feasible.

Formal Experimental Designs

- Incorporate randomization, control groups, and systematic procedures.
- Aim to minimize bias and establish causality with greater validity.

Informal Experimental Designs

Before-After without Control

- Single group measured before and after treatment
- Simplest design but limited validity

Test area:	Level of phenomenon before treatment (X)	Treatment introduced $\longrightarrow$	Level of phenomenon after treatment (Y)
	Treatment Effect = (Y) – (X)		

- A single test group is selected and the dependent variable is measured before the treatment (X).
- The treatment is introduced and the dependent variable is measured again after the treatment (Y).
- Treatment effect is calculated as $(Y - X)$, but results may be affected by extraneous changes over time.

Informal Experimental Designs

Before-After without Control

Advantages

- Simple to conduct and requires only a single test group.
- Useful when a control group cannot be formed.

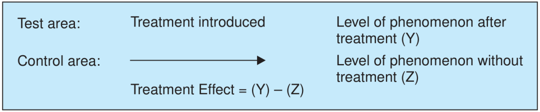
Disadvantages

- Treatment effect may be influenced by extraneous factors over time.
- Absence of a control group weakens causal conclusions.

Informal Experimental Designs

After-Only with Control

- Two groups measured only after treatment
- Control and experimental groups compared
- **Definition:** A control group is a group in an experiment that does not receive the intervention or treatment being tested. It is used as a baseline to compare and interpret the effect of the intervention.



- Two groups are selected: a test area (receives treatment) and a control area (no treatment).
- The dependent variable is measured in both areas at the same time after treatment.
- Treatment effect is calculated as $(Y - Z)$, assuming both areas are identical except for the treatment.

Informal Experimental Designs

After-Only with Control

Advantages

- Avoids problems due to passage of time since measurement is taken only after treatment.
- Presence of a control group improves comparison over uncontrolled designs.

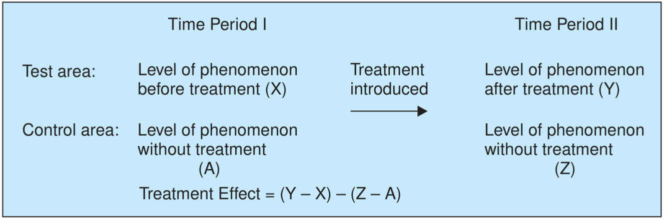
Disadvantages

- Assumes test and control groups are identical, which may not always be true.
- Extraneous differences between groups may affect the treatment effect.

Informal Experimental Designs

Before-After with Control

- Both groups measured before and after treatment
- Most rigorous informal design



- Two areas (test and control) are observed for an identical period before the treatment.
- Treatment is applied only to the test area and both areas are observed again after treatment.
- Treatment effect is calculated as $(Y - X) - (Z - A)$, minimizing time and group-related extraneous effects.

Informal Experimental Designs

Before-After with Control

Advantages

- Controls for extraneous variations due to time and group differences.
- Provides a more accurate estimate of treatment effect than informal designs.

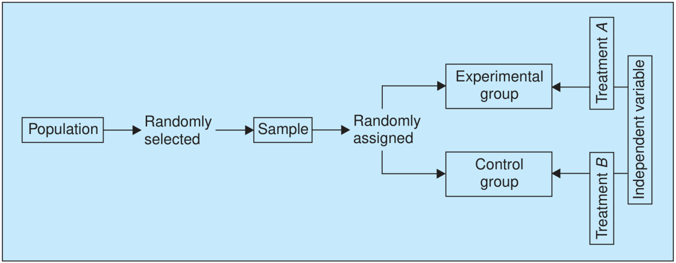
Disadvantages

- Requires comparable control and test groups, which may be difficult to obtain.
- Needs data collection over two time periods, increasing time and cost.

Formal Experimental Designs

Completely Randomized Design

- Random assignment of experimental units to treatments
- Simple and powerful design



- Based on randomization; subjects are randomly assigned to treatments.
- It is the simplest experimental design and is usually analysed using one-way ANOVA.
- Suitable when experimental units are homogeneous and all uncontrolled variations are treated as chance error.

Formal Experimental Designs

Completely Randomized Design

Advantages

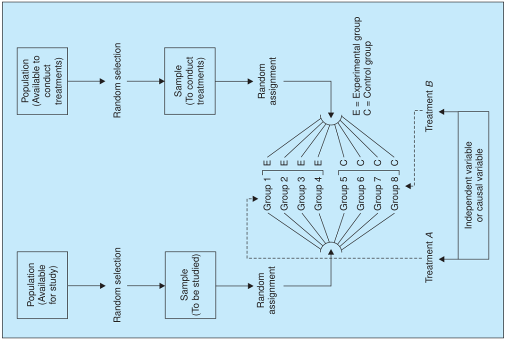
- Simple to design and analyse; randomization provides maximum degrees of freedom for error.
- Suitable when experimental units are homogeneous and extraneous variations are minimal.

Disadvantages

- Not effective when experimental units are heterogeneous.
- Does not provide control over known extraneous variables.

Formal Experimental Designs

Random Replications Design



- Uses multiple replications for each treatment to reduce the effect of uncontrolled extraneous variables.
- Each replication involves random selection of subjects and random assignment to experimental and control groups.
- The design controls extraneous effects and randomizes individual differences, extending the two-group randomized design.

Formal Experimental Designs

Random Replications Design

Advantages

- Reduces the effect of extraneous variables by using multiple replications for each treatment.
- Randomizes individual differences, leading to more reliable experimental results.

Disadvantages

- Requires a larger sample size, increasing time and cost.
- More complex to administer than a simple randomized design.

Formal Experimental Designs

Randomized Block Design

- Subjects grouped by relevant characteristics
- Controls for known sources of variation

	Very low I.Q.	Low I.Q.	Average I.Q.	High I.Q.	Very high I.Q.
	Student A	Student B	Student C	Student D	Student E
Form 1	82	67	57	71	73
Form 2	90	68	54	70	81
Form 3	86	73	51	69	84
Form 4	93	77	60	65	71

- Subjects are first divided into homogeneous groups called blocks based on a variable related to the response.
- Within each block, subjects are randomly assigned so that each treatment appears the same number of times.
- This design applies local control of extraneous variables and is analysed using two-way ANOVA.

Formal Experimental Designs

Randomized Block Design

Advantages

- Controls known extraneous variables by grouping experimental units into homogeneous blocks.
- Increases precision of treatment comparison compared to completely randomized design.

Disadvantages

- Requires identification of an appropriate blocking variable.
- Becomes complex and less efficient if blocks are not truly homogeneous.

Formal Experimental Designs

Latin Square Design

- Controls for two sources of variation simultaneously
- More complex but more efficient

FERTILITY LEVEL

Seeds differences

	I	II	III	IV	V
X_1	A	B	C	D	E
X_2	B	C	D	E	A
X_3	C	D	E	A	B
X_4	D	E	A	B	C
X_5	E	A	B	C	D

- Used when two major extraneous factors exist; treatments are arranged so each appears once in every row and column.
- Rows and columns act as two blocking factors, controlling variability due to these extraneous sources.
- Analysis is similar to two-way ANOVA, assuming no interaction between treatments and blocking factors.

Formal Experimental Designs

Latin Square Design

Advantages

- Controls two major extraneous variables simultaneously through row and column blocking.
- More efficient than C.R. and R.B. designs when two sources of variability are present.

Disadvantages

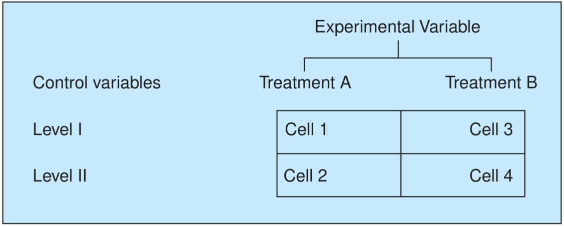
- Assumes no interaction between treatments and blocking factors.
- Requires equal number of rows, columns, and treatments, limiting flexibility.

Formal Experimental Designs

Factorial Design

- Studies combined effects of multiple factors
- Examines interactions between variables

2 × 2 SIMPLE FACTORIAL DESIGN



- Involves two experimental treatments and two levels of a control variable, forming four experimental cells.
- Subjects are randomly assigned to each cell, and each cell represents a unique treatment–level combination.
- The design allows study of main effects of treatments and levels, as well as their interaction effects.

Formal Experimental Designs

Factorial Design

Advantages

- Controls two major extraneous variables simultaneously through row and column blocking.
- More efficient than C.R. and R.B. designs when two sources of variability are present.

Disadvantages

- Assumes no interaction between treatments and blocking factors.
- Requires equal number of rows, columns, and treatments, limiting flexibility.

Choosing the Right Research Design

Pick Your Design Based on Your Goal:

- **Exploration**: Use a flexible plan so you can look at many different ideas
- **Description**: Use a strict, structured plan to get accurate and reliable results
- **Diagnosis**: Clearly define what you are studying and how things are related
- **Experimentation**: Use a design that lets you test if one thing causes another

Key Point: The best design is the one that fits your research goal.

Four Main Parts of Research Design

The Four Key Parts Are:

1. **Sampling Design**
   - How you choose the people or items to study
   - Decides who is included and how many you need
2. **Observational Design**
   - The conditions and rules for making your observations
   - Helps control for outside factors that could affect your results
3. **Statistical Design**
   - How many things you will observe
   - The ways you will analyze your data and test your ideas
4. **Operational Design**
   - The step-by-step methods for carrying out your research
   - Makes sure everything is done the same way each time

Basic Requirements for a Good Research Design

What You Must Include:

1. **Clear problem statement**: Clearly say what you are trying to find out
2. **Procedures and techniques**: Explain how you will collect your information
3. **Population definition**: Clearly describe who or what you are studying
4. **Analysis methods**: Say how you will look at and understand your data
5. **Validity and reliability**: Make sure your results are accurate and not biased

Note: Leaving out any of these can make your research less trustworthy or useful.

Key Takeaways

- **Research design** is like a plan that helps you collect and analyze data in the best way.
- **Why it matters**: A good design reduces mistakes, gives trustworthy results, and saves time and resources.
- **Good design**: Should be flexible when exploring new ideas, but strict when describing things clearly.
- **Four main parts**: Sampling, observation, statistics, and how you carry out the research.
- **Type of design**: Should match your research goal—whether you're exploring, describing, diagnosing, or experimenting.
- **Experimental vs. non-experimental**: Choose based on whether you can change (control) the variables or just observe them.
- **Simple vs. formal experiments**: Simple designs are easier, but formal ones give you more control.
- **Good planning**: Is necessary to get results that are accurate and reliable.

Summary: Steps for Preparing a Research Design

1. **Decide your research goal**
   - Are you exploring, describing, diagnosing, or experimenting?
2. **Choose the right design**
   - Pick a design that fits your goal
3. **Plan the main parts**
   - Think about how you will sample, observe, analyze, and carry out the research
4. **Think about real-world limits**
   - Consider your time, budget, team skills, and resources
5. **Add quality checks**
   - Reduce mistakes, make results reliable, and control outside influences

Questions and Discussion

Critical Question to Ask

"Does my research design minimize bias, control extraneous variables, and provide maximal information efficiency?"

Thank you!